



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

HEXAGONAL ARCHITECTURE PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Architecture

TOTAL: 10 QUESTIONS

Q1 What is Hexagonal Architecture and what problem in Threat Model Architecture does it solve?

Q6 How does Hexagonal Architecture handle reliability and high availability?

Q2 What are the core components or concepts in Hexagonal Architecture?

Q7 What observability does Hexagonal Architecture provide out of the box?

Q3 How does Hexagonal Architecture compare to its main configuration alternatives?

Q8 What are common performance bottlenecks in Hexagonal Architecture?

Q4 How do you get started with Hexagonal Architecture for a new project?

Q9 What security best practices apply when running Hexagonal Architecture?

Q5 What configuration options most affect Hexagonal Architecture's behavior in production?

Q10 What mistakes do teams commonly make when adopting Hexagonal Architecture?



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Hexagonal Architecture is a tool or platform

Mitigation

Hexagonal Architecture is a tool or platform that automates or simplifies a specific concern in the Architecture domain, reducing manual effort and operational complexity for engineering teams.

A6 Hexagonal Architecture provides HA through leader election, replication, or stateless horizontal scaling.

Availability

Hexagonal Architecture provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Hexagonal Architecture has key building blocks that

Primitives

Hexagonal Architecture has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Hexagonal Architecture exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces.

Automation

Hexagonal Architecture exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 Hexagonal Architecture trades off simplicity against feature richness

Feature

Hexagonal Architecture trades off simplicity against feature richness differently than its alternatives. Choose Hexagonal Architecture when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches.

Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install Hexagonal Architecture via its package manager or binary release.

Gotchas

Install Hexagonal Architecture via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Hexagonal Architecture with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

Assessment

Run Hexagonal Architecture with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels.

Configuration

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.

Documentation

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.